

Day 54 (3) - VRF

Thursday, September 17, 2026

9:10 AM

VRF Info

Virtual Routing & Forwarding (VRF) is used to divide a single router into multiple virtual routers. Similar to how VLANs are used to divide a single switch (LAN) into multiple virtual switches (VLANs)

- **VRF** works by allowing a router to build multiple separate routing tables.
 - Interfaces (*Layer 3* only) and routes are configured to be in a specific **VRF** (**VRF** instance).
 - Only *router interfaces*, **SVIs**, and *routed ports* on **multilayer switches** can be configured in a **VRF**.
- Traffic in one **VRF** cannot be forwarded out of an interface in another **VRF**.
 - However, **VRF Leaking** can be configured to allow traffic to pass between **VRF**'s
- **VRF** is commonly used by service providers to allow one device to carry traffic from multiple customers.
 - Each customer's traffic is isolated from the others.
 - Customer IP addresses can overlap without issues.

VRF is commonly used in MPLS, but for the CCNA, the kind of VRF we're talking about is VRF-lite. (which VRF without MPLS)

VRF Configuration

- *show ip vrf*
 - Shows all configured **VRF**s on the router.
- *show ip route vrf VRF-NAME*
 - Shows the IP routing table for a specific **VRF**. (**VRF** routing tables do not show up in the normal *show ip route* command.)
- *ip vrf VRF-NAME*
 - Creates a new **VRF** on the router.
- *ip vrf forwarding VRF-NAME*
 - Applies the specified **VRF** onto the current interface.
- *ping vrf VRF-NAME IP-ADDRESS*
 - Pings an IP address in the specified **VRF**.